Reg. No. 7113823020 12

SRI RAMAKRISHNA INSTITUTE OF TECHNOLOGY

[Educational Service: SNR Sons Charitable Trust]
[Autonomous Institution, Reaccredited by NAAC with 'A' Grade]
[Approved by AICTE New Delhi, Permanently Affiliated to Anna University, Chennai]

Department of Computer Science and Engineering

Internal Test – II					
Date	**13.10.2025**	**Department**	**CSE**		
Semester	**V**	**Class/section**	**III CSE**		
Duration	**2:00 Hours**	**Maximum marks**	**50**	**Regulation**	**R2020**
Course Code &Title:		**20CSP11 MOBILE NETWORKS**			

Course Outcomes Addressed:
CO1: Understand challenges in routing in different kinds of networks.
CO2: Explore the features of the protocols used in networks.
CO3: Analyze the issues of different protocols for different kinds of networks.

S. No	Questions	Cognitive Level/CO
	PART – A (Answer all questions) (5*2 =10 Marks)	
1.	What are the main goals of Mobile IP?	C1/CO1
2.	Define Tunneling in Mobile IP.	C1/CO2
3.	List the main drawbacks of Traditional TCP in wireless environments.	C2/CO1
4.	Write any two features of WAP 2.0.	C1/CO2
5.	Define a Mobile Ad-hoc Network (MANET).	C2/CO1
	PART – B (Answer any three questions) (3*8 =24 Marks)	**Cognitive Level/CO**
6.	Explain the Agent Advertisement and Discovery process in Mobile IP with a diagram.	C3/CO2
7.	Describe Mobile TCP and explain how it helps during handoff.	C3/CO1
8.	Compare WAP 1.x and WAP 2.0 architectures in terms of protocol stack, efficiency, and limitations.	C4/CO2
9.	A telecom provider wants to support seamless mobility. Analyze how DSDV and DSR protocols handle dynamic topology changes in MANETs. Which one is more suitable for highly mobile users? Justify.	C4/CO3
	PART – C (Answer any one question) (1*16 =16 Marks)	
10.	A mobile user moves between different foreign networks while maintaining an active session. Explain how Mobile IP handles agent discovery, registration, tunneling, and route optimization to ensure seamless communication. Use an architecture diagram to support your explanation. (Adapted from GATE CS 2015, Question on Mobile IP and Tunneling)	C3/CO1

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11.	In a wireless sensor network for smart healthcare monitoring, nodes have limited energy and storage. (a) Compare the following wireless protocols: Bluetooth (IEEE 802.15.1), ZigBee (IEEE 802.15.4), IrDA, Wi-Fi (IEEE 802.11b) with respect to: Power consumption, Range, Data rate, Scalability. (b) Based on your comparison, recommend the most suitable protocol for healthcare monitoring applications. Justify your answer. (Adapted from GATE 2019 CS/IT, Wireless Networks)	C4/CO3

S. No.	Cognitive Level addressed	Marks	Percentage
1.	Remember (C1)	06	8%
2.	Understand (C2)	04	5%
3.	Apply (C3)	32	43%
4.	Analyze (C4)	32	43%
5.	Evaluate (C5)	-	-
6.	Create (C6)	-	-
Total		74	100

Course Instructor

Program Assessment Committee

26/9

Program Coordinator